

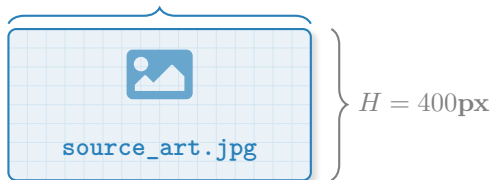
The Algorithmic Ceiling of max-width: 100%

Mathematical refusal to stretch beyond physical source resolution in flexible media

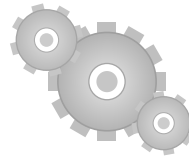
1. Intrinsic Source Data

The physical image file has absolute pixel dimensions encoded in its raw bitmap matrix.

$$W_{\text{intrinsic}} = 800\text{px}$$



2. Browser Algorithm

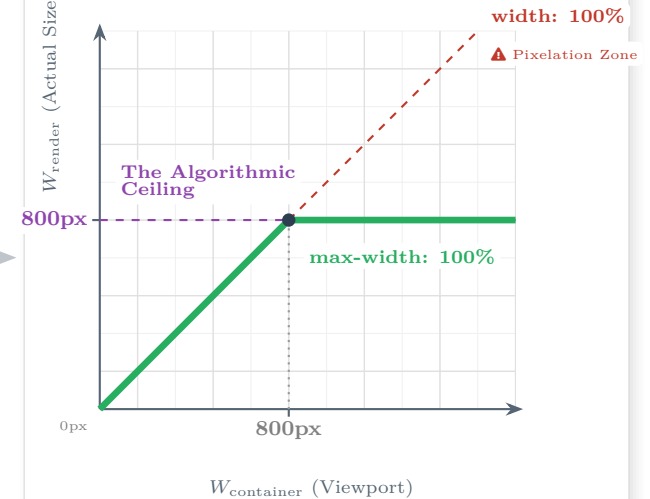


`max-width: 100%`

Computes a strict bounding function against the parent container width.

$$W_{\text{render}} = \min(W_{\text{intrinsic}}, W_{\text{container}})$$

3. Visualizing the Math



CSS Implementation

```
style.css
img {
  max-width: 100%; /* Ceiling */
  height: auto; /* Scale Lock */
  display: block;
}
```

The Structural Engine: The browser continuously calculates the bounding box of the parent HTML element. If the viewport shrinks, 100% dictates the image must algorithmically scale down to fit, triggering `height: auto` to mathematically preserve the original 2:1 aspect ratio. Conversely, on wide screens, the `max` function prohibits the browser from interpolating artificial pixels, refusing to stretch past its native source matrix (800px), guaranteeing absolute art direction fidelity.